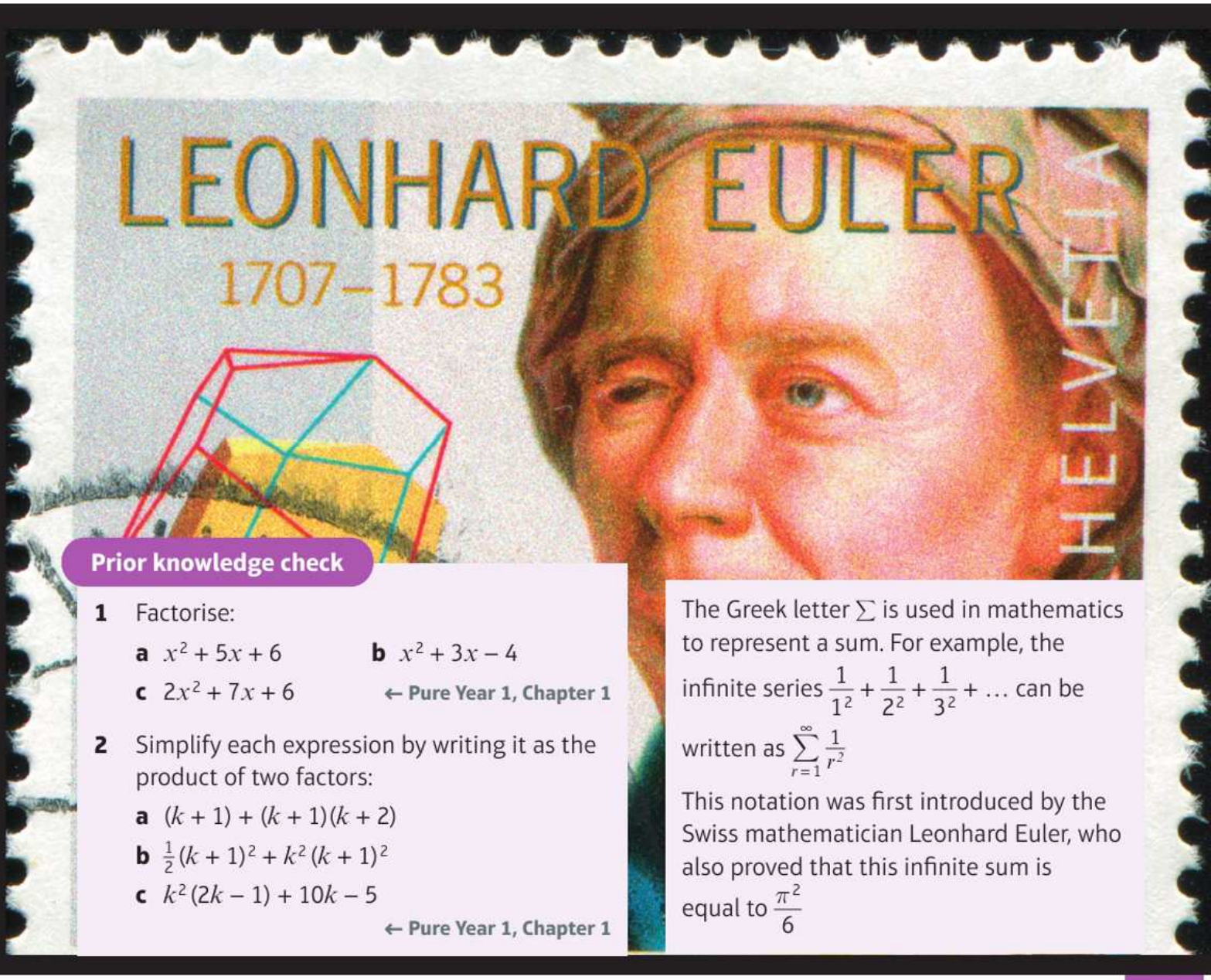
Series

3

Objectives

After completing this chapter you should be able to:

- Use standard results for $\sum_{r=1}^{n} 1$ and $\sum_{r=1}^{n} r$ → pages 44–47
- Use standard results for $\sum_{r=1}^{n} r^2$ and $\sum_{r=1}^{n} r^3$ → pages 47–51
- Evaluate and simplify series of the form $\sum_{r=m}^{n} \mathrm{f}(r)$, where $\mathrm{f}(r)$ is linear, quadratic or cubic → pages 44–51



Prior knowledge check

1 Factorise:

a $x^2 + 5x + 6$ **b** $x^2 + 3x - 4$

c $2x^2 + 7x + 6$ ← Pure Year 1, Chapter 1

2 Simplify each expression by writing it as the product of two factors:

a $(k + 1) + (k + 1)(k + 2)$

b $\frac{1}{2}(k + 1)^2 + k^2(k + 1)^2$

c $k^2(2k - 1) + 10k - 5$

← Pure Year 1, Chapter 1

The Greek letter $\sum$ is used in mathematics to represent a sum. For example, the infinite series $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \ldots$ can be written as $\sum_{r=1}^{\infty} \frac{1}{r^2}$

This notation was first introduced by the Swiss mathematician Leonhard Euler, who also proved that this infinite sum is equal to $\frac{\pi^2}{6}$

3.1 Sums of natural numbers

You can use **sigma notation** to write series clearly and concisely. For example:

$$\sum_{r=1}^{3}(10r-1) = (10\times 1-1)+(10\times 2-1)+(10\times 3-1)$$
$$= 9+19+29 = 57$$
$$\sum_{r=1}^{n} r^2 = 1^2+2^2+3^2+\ldots+n^2$$

Links A series is the **sum** of the terms in a sequence.

← Pure Year 2, Section 3.2

Notation The numbers below and above the $\sum$ tell you which value of r to begin at, and which value to end at. You go up in increments of 1 each time.

- **To find the sum of a series of constant terms you can use the formula $\sum_{r=1}^{n} 1 = n$.**
- **The formula for the sum of the first n natural numbers is $\sum_{r=1}^{n} r = \frac{1}{2}n(n+1)$.**

Notation $\sum_{r=1}^{n} r = 1+2+3+\ldots+n$

Example 1

Evaluate: **a** $\sum_{r=1}^{4}(2r-1)$ **b** $\sum_{r=1}^{50} r$ **c** $\sum_{r=21}^{50} r$

a $\sum_{r=1}^{4}(2r-1) = (2\times 1-1)+(2\times 2-1)+(2\times 3-1)+(2\times 4-1)$
$= 1+3+5+7$
$= 16$

There are only 4 terms in this series. Write out each one then find the sum.

b $\sum_{r=1}^{50} r = \frac{1}{2}\times 50\times 51 = 1275$

Substitute $n = 50$ in $\sum_{r=1}^{n} r = \frac{1}{2}n(n+1)$.

c $\sum_{r=21}^{50} r = \sum_{r=1}^{50} r - \sum_{r=1}^{20} r = 1275 - \frac{1}{2}\times 20\times 21$
$= 1275 - 210 = 1065$

Problem-solving

$\sum_{r=21}^{50} r = 21+22+23+\ldots+49+50$

Find the sum of the natural numbers up to 50, then subtract the sum of the natural numbers up to 20.

- **To find the sum of a series that does not start at $r = 1$, use $\sum_{r=k}^{n} f(r) = \sum_{r=1}^{n} f(r) - \sum_{r=1}^{k-1} f(r)$.**

Watch out You need to subtract the sum up to $k-1$, not k.

Example 2

Show that $\sum_{r=5}^{2N-1} r = 2N^2 - N - 10$, $N \geqslant 3$.

$$\begin{aligned}\sum_{r=5}^{2N-1} r &= \sum_{r=1}^{2N-1} r - \sum_{r=1}^{4} r \\ &= \tfrac{1}{2} \times (2N-1)((2N-1)+1) - \tfrac{1}{2} \times 4 \times 5 \\ &= \tfrac{1}{2} \times (2N-1)(2N) - 10 \\ &= \tfrac{1}{2} \times (4N^2 - 2N) - 10 \\ &= 2N^2 - N - 10\end{aligned}$$

Substitute $n = 2N - 1$ and $n = 4$ in $\frac{1}{2}n(n+1)$.

You can rearrange expressions involving sigma notation. This allows you to evaluate the sums of more complicated series.

- $\sum_{r=1}^{n} k\mathbf{f}(r) = k \sum_{r=1}^{n} \mathbf{f}(r)$
- $\sum_{r=1}^{n} (\mathbf{f}(r) + \mathbf{g}(r)) = \sum_{r=1}^{n} \mathbf{f}(r) + \sum_{r=1}^{n} \mathbf{g}(r)$

Example 3

Evaluate $\sum_{r=1}^{25} (3r + 1)$

$$\begin{aligned}\sum_{r=1}^{25} (3r+1) &= 3\sum_{r=1}^{25} r + \sum_{r=1}^{25} 1 \\ &= 3 \times \tfrac{1}{2} \times 25 \times 26 + 25 \\ &= 975 + 25 = 1000\end{aligned}$$

Use the rules given above to write the expression in terms of $\sum_{r=1}^{25} r$ and $\sum_{r=1}^{25} 1$.

Example 4

a Show that $\sum_{r=1}^{n} (7r - 4) = \frac{1}{2}n(7n - 1)$.

b Hence evaluate $\sum_{r=20}^{50} (7r - 4)$.

a
$$\begin{aligned}\sum_{r=1}^{n} (7r-4) &= 7\sum_{r=1}^{n} r - 4\sum_{r=1}^{n} 1 \\ &= \tfrac{7}{2}n(n+1) - 4n \\ &= \tfrac{1}{2}n(7n + 7 - 8) \\ &= \tfrac{1}{2}n(7n - 1)\end{aligned}$$

$7\sum_{r=1}^{n} r = 7 \times \frac{1}{2}n(n+1)$ and $4\sum_{r=1}^{n} 1 = 4 \times n$

b
$$\begin{aligned}\sum_{r=20}^{50} (7r-4) &= \sum_{r=1}^{50} (7r-4) - \sum_{r=1}^{19} (7r-4) \\ &= \tfrac{1}{2} \times 50 \times 349 - \tfrac{1}{2} \times 19 \times 132 \\ &= 8725 - 1254 \\ &= 7471\end{aligned}$$

Use the result from part **a** to find each sum quickly.
$7 \times 50 - 1 = 349$ and $7 \times 19 - 1 = 132$

Exercise 3A

1 Evaluate:

a $\sum_{r=0}^{3}(2r+1)$ **b** $\sum_{r=1}^{40}r$ **c** $\sum_{r=1}^{20}r$ **d** $\sum_{r=1}^{99}r$

e $\sum_{r=10}^{40}r$ **f** $\sum_{r=100}^{200}r$ **g** $\sum_{r=21}^{40}r$ **h** $\sum_{r=1}^{k}r+\sum_{r=k+1}^{80}r,\ 0<k<80$

E/P **2** Given that $\sum_{r=1}^{n}r=528$, find the value of n. **(4 marks)**

E/P **3** Given that $\sum_{r=1}^{k}r=\frac{1}{2}\sum_{r=1}^{20}r$, find the value of k. **(4 marks)**

E **4 a** Find an expression for $\sum_{r=1}^{2n-1}r$. **(3 marks)**

b Hence show that $\sum_{r=n+1}^{2n-1}r=\frac{3}{2}n(n-1),\ n\geqslant 2$. **(3 marks)**

E **5** Show that $\sum_{r=n-1}^{2n}r=\frac{1}{2}(n+2)(3n-1),\ n\geqslant 1$. **(5 marks)**

6 a Show that $\sum_{r=1}^{n^2}r-\sum_{r=1}^{n}r=\frac{1}{2}n(n^3-1)$.

b Hence evaluate $\sum_{r=10}^{81}r$.

Hint Use your result from part **a**.

7 Calculate the sum of each series:

a $\sum_{r=1}^{55}(3r-1)$ **b** $\sum_{r=1}^{90}(2-7r)$ **c** $\sum_{r=1}^{46}(9+2r)$

8 Show that:

a $\sum_{r=1}^{n}(3r+2)=\frac{1}{2}n(3n+7)$ **b** $\sum_{r=1}^{2n}(5r-4)=n(10n-3)$

c $\sum_{r=1}^{n+2}(2r+3)=(n+2)(n+6)$ **d** $\sum_{r=3}^{n}(4r+5)=(2n+11)(n-2)$

E/P **9 a** Show that $\sum_{r=1}^{k}(4r-5)=2k^2-3k$. **(5 marks)**

b Find the smallest value of k for which $\sum_{r=1}^{k}(4r-5)>4850$. **(4 marks)**

E/P **10** Given that $f(r)=ar+b$ and $\sum_{r=1}^{n}f(r)=\frac{1}{2}n(7n+1)$, find the constants a and b. **(5 marks)**

E **11 a** Show that $\sum_{r=1}^{4n-1}(3r+1)=24n^2-2n-1,\ n\geqslant 1$. **(5 marks)**

b Hence calculate $\sum_{r=1}^{99}(3r+1)$. **(2 marks)**

E **12 a** Show that $\sum_{r=1}^{2k+1}(4-5r)=-(2k+1)(5k+1),\ k\geqslant 0$. **(5 marks)**

b Hence evaluate $\sum_{r=1}^{25}(4-5r)$. **(2 marks)**

c Find the value of $\sum_{r=1}^{15}(5r-4)$. **(1 mark)**

(P) **13** Given that $\sum_{r=1}^{n} f(r) = n^2 + 4n$, deduce an expression for $f(r)$ in terms of r.

(E/P) **14** $f(r) = ar + b$, where a and b are rational constants.

Given that $\sum_{r=1}^{4} f(r) = 36$ and $\sum_{r=1}^{6} f(r) = 78$,

a find an expression for $\sum_{r=1}^{n} f(r)$ **(6 marks)**

b hence calculate $\sum_{r=1}^{10} f(r)$. **(2 marks)**

Challenge

Given that $\sum_{r=n}^{2n} (12 - 2r) = 0$, find the value of n.

3.2 Sums of squares and cubes

The expression for $\sum_{r=1}^{n} 1$ is linear, and the expression for $\sum_{r=1}^{n} r$ is quadratic. Similarly, you can find a cubic expression for the sum of the **squares** of the first n natural numbers, and a quartic expression for the sum of the **cubes** of the first n natural numbers.

- **The formula for the sum of the squares of the first n natural numbers is $\sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1)$.**
- **The formula for the sum of the cubes of the first n natural numbers is $\sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n+1)^2$.**

Links You can prove both of these results using **mathematical induction**. → Section 8.1

Example 5

Evaluate: **a** $\sum_{r=20}^{40} r^2$ **b** $\sum_{r=1}^{25} r^3$

a $\sum_{r=20}^{40} r^2 = \sum_{r=1}^{40} r^2 - \sum_{r=1}^{19} r^2$

$= \frac{1}{6} \times 40(40+1)(80+1) - \frac{1}{6} \times 19(19+1)(38+1)$

$= 22\,140 - 2470 = 19\,670$

Substitute $n = 40$ and $n = 19$ in $\sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1)$

b $\sum_{r=1}^{25} r^3 = \frac{1}{4} \times 25^2 \times 26^2 = 105\,625$

Substitute $n = 25$ in $\frac{1}{4}n^2(n+1)^2$.

Example 6

a Show that $\sum_{r=n+1}^{2n} r^2 = \frac{1}{6}n(2n+1)(7n+1)$.

b Verify that the result is true for $n = 1$ and $n = 2$.

a $\sum_{r=n+1}^{2n} r^2 = \sum_{r=1}^{2n} r^2 - \sum_{r=1}^{n} r^2$

$= \frac{1}{6} \times 2n(2n + 1)(4n + 1) - \frac{1}{6}n(n + 1)(2n + 1)$

$= \frac{1}{6}n(2n + 1)(2(4n + 1) - (n + 1))$

$= \frac{1}{6}n(2n + 1)(7n + 1)$

b When $n = 1$:

$\sum_{r=n+1}^{2n} r^2 = \sum_{r=2}^{2} r^2 = 2^2 = 4$

$\frac{1}{6}n(2n + 1)(7n + 1) = \frac{1}{6} \times 3 \times 8 = 4$ ✓

When $n = 2$:

$\sum_{r=n+1}^{2n} r^2 = \sum_{r=3}^{4} r^2 = 3^2 + 4^2 = 25$

$\frac{1}{6}n(2n + 1)(7n + 1) = \frac{2}{6} \times 5 \times 15 = 25$ ✓

Replace n by $2n$ in $\frac{1}{6}n(n + 1)(2n + 1)$.

Problem-solving

Look for common factors in each part of the expression. Here you can take out a factor of $\frac{1}{6}n(2n + 1)$.

Watch out When you have been asked to find a general result for a sum it is good practice to test it for small values of n. It will not prove that you are correct, but if one value of n does not work, you know that your result is incorrect.

Example 7

a Show that $\sum_{r=1}^{n}(r^2 + r - 2) = \frac{1}{3}n(n + 4)(n - 1)$.

b Hence find the sum of the series $4 + 10 + 18 + 28 + 40 + \ldots + 418$.

a $\sum_{r=1}^{n}(r^2 + r - 2)$

$= \sum_{r=1}^{n} r^2 + \sum_{r=1}^{n} r - 2\sum_{r=1}^{n} 1$

$= \frac{1}{6}n(n + 1)(2n + 1) + \frac{1}{2}n(n + 1) - 2n$

$= \frac{1}{6}n((n + 1)(2n + 1) + 3(n + 1) - 12)$

$= \frac{1}{6}n(2n^2 + 3n + 1 + 3n + 3 - 12)$

$= \frac{1}{6}n(2n^2 + 6n - 8)$

$= \frac{1}{3}n(n^2 + 3n - 4)$

$= \frac{1}{3}n(n + 4)(n - 1)$

b $0 + 4 + 10 + 18 + 28 + 40 + \ldots + 418$

$= \sum_{r=1}^{20}(r^2 + r - 2)$

$= \frac{1}{3} \times 20(20 + 4)(20 - 1)$

$= 3040$

Use the results for $\sum_{r=1}^{n} r^2$, $\sum_{r=1}^{n} r$ and $\sum_{r=1}^{n} 1$.

Problem-solving

The question says 'hence' so use your answer to part **a**. When $r = 1$, $r^2 + r - 2 = 0$, and when $r = 20$, $r^2 + r - 2 = 418$, so you can write the sum as $\sum_{r=1}^{20}(r^2 + r - 2)$.

Example 8

a Show that $\sum_{r=1}^{n} r(r+3)(2r-1) = \frac{1}{6}n(n+1)(3n^2+an+b)$, where a and b are integers to be found.

b Hence calculate $\sum_{r=11}^{40} r(r+3)(2r-1)$.

a $\sum_{r=1}^{n} r(r+3)(2r-1)$

$= \sum_{r=1}^{n} (2r^3 + 5r^2 - 3r)$

$= 2\sum_{r=1}^{n} r^3 + 5\sum_{r=1}^{n} r^2 - 3\sum_{r=1}^{n} r$

$= \frac{2}{4}n^2(n+1)^2 + \frac{5}{6}n(n+1)(2n+1) - \frac{3}{2}n(n+1)$

$= \frac{1}{6}n(n+1)(3n(n+1) + 5(2n+1) - 9)$

$= \frac{1}{6}n(n+1)(3n^2 + 13n - 4)$

b $\sum_{r=11}^{40} r(r+3)(2r-1)$

$= \sum_{r=1}^{40} r(r+3)(2r-1) - \sum_{r=1}^{10} r(r+3)(2r-1)$

$= \frac{1}{6}(40 \times 41 \times 5316) - \frac{1}{6}(10 \times 11 \times 426)$

$= 1\,453\,040 - 7810$

$= 1\,445\,230$

First multiply out the brackets.

Use these rules:

$\sum_{r=1}^{n} kf(r) = k\sum_{r=1}^{n} f(r)$

$\sum_{r=1}^{n} (f(r) + g(r)) = \sum_{r=1}^{n} f(r) + \sum_{r=1}^{n} g(r)$

Use the results for $\sum_{r=1}^{n} r^3$, $\sum_{r=1}^{n} r^2$ and $\sum_{r=1}^{n} r$.

Substitute $n = 40$ and $n = 10$ in the result for **a**.

Exercise 3B

1 Evaluate:

a $\sum_{r=1}^{4} r^2$ **b** $\sum_{r=1}^{40} r^2$ **c** $\sum_{r=21}^{40} r^2$ **d** $\sum_{r=1}^{99} r^3$

e $\sum_{r=1}^{100} r^3$ **f** $\sum_{r=100}^{200} r^3$ **g** $\sum_{r=1}^{k} r^2 + \sum_{r=k+1}^{80} r^2$, $0 < k < 80$.

2 Show that:

a $\sum_{r=1}^{2n} r^2 = \frac{1}{3}n(2n+1)(4n+1)$ **b** $\sum_{r=1}^{2n-1} r^2 = \frac{1}{3}n(2n-1)(4n-1)$

c $\sum_{r=n}^{2n} r^2 = \frac{1}{6}n(n+1)(14n+1)$

(P) **3** Show that, for any $k \in \mathbb{N}$, $\sum_{r=1}^{n+k} r^3 = \frac{1}{4}(n+k)^2(n+k+1)^2$

(E) **4 a** Show that $\sum_{r=n+1}^{3n} r^3 = n^2(4n+1)(5n+2)$ **(3 marks)**

b Hence evaluate $\sum_{r=11}^{30} r^3$. **(2 marks)**

E **5 a** Show that $\sum_{r=n}^{2n} r^3 = \frac{3}{4}n^2(n+1)(5n+1)$. **(3 marks)**

b Hence evaluate $\sum_{r=30}^{60} r^3$. **(2 marks)**

6 Evaluate:

a $\sum_{m=1}^{30} (m^2 - 1)$ **b** $\sum_{r=1}^{40} r(r+4)$ **c** $\sum_{r=1}^{80} r(r^2+3)$ **d** $\sum_{r=11}^{35} (r^3 - 2)$

E **7 a** Show that $\sum_{r=1}^{n} (r+2)(r+5) = \frac{1}{3}n(n^2 + 12n + 41)$ **(4 marks)**

b Hence calculate $\sum_{r=10}^{50} (r+2)(r+5)$. **(3 marks)**

E **8 a** Show that $\sum_{r=1}^{n} (r^2 + 3r + 1) = \frac{1}{3}n(n+a)(n+b)$, where a and b are integers to be found. **(4 marks)**

b Hence evaluate $\sum_{r=19}^{40} (r^2 + 3r + 1)$. **(3 marks)**

E **9 a** Show that $\sum_{r=1}^{n} r^2(r-1) = \frac{1}{12}n(n+1)(3n^2 - n - 2)$. **(4 marks)**

b Hence show that $\sum_{r=1}^{2n-1} r^2(r-1) = \frac{1}{3}n(2n-1)(6n^2 - 7n + 1)$ **(4 marks)**

E **10 a** Show that $\sum_{r=1}^{n} (r+1)(r+3) = \frac{1}{6}n(2n^2 + an + b)$, where a and b are integers to be found. **(4 marks)**

b Hence find an expression, only in terms of n, for $\sum_{r=n+1}^{2n} (r+1)(r+3)$. **(3 marks)**

E **11 a** Show that $\sum_{r=1}^{n} (r+3)(r+4) = \frac{1}{3}n(n^2 + an + b)$, where a and b are integers to be found. **(4 marks)**

b Hence find an expression, only in terms of n, for $\sum_{r=n+1}^{3n} (r+3)(r+4)$. **(3 marks)**

E **12 a** Show that $\sum_{r=1}^{n} r(r+3)^2 = \frac{1}{4}n(n+1)(n^2 + an + b)$, where a and b are integers to be found. **(5 marks)**

b Hence evaluate $\sum_{r=10}^{20} r(r+3)^2$. **(3 marks)**

P **13 a** Show that, for any $k \in \mathbb{N}$, $\sum_{r=1}^{kn} (2r-1) = k^2n^2$.

b Hence find a value of n such that $\sum_{r=1}^{5n} (2r-1) = \sum_{r=1}^{n} r^3$.

E/P **14 a** Show that $\sum_{r=1}^{n}(r^3 - r^2) = \frac{1}{12}n(n+1)(n-1)(3n+2)$. **(4 marks)**

b Hence find the value of n that satisfies $\sum_{r=1}^{n}(r^3 - r^2) = \sum_{r=1}^{n}7r$. **(5 marks)**

Challenge

a Find polynomials $f_2(x)$, $f_3(x)$, $f_4(x)$ such that for every $n \in \mathbb{N}$:

$$\sum_{r=1}^{n}f_2(r) = n^2, \quad \sum_{r=1}^{n}f_3(r) = n^3, \quad \sum_{r=1}^{n}f_4(r) = n^4$$

b Hence show that for any linear, quadratic, or cubic polynomial $h(x)$ there exists a polynomial $g(x)$ such that $\sum_{r=1}^{n}g(r) = n(h(n))$.

Hint The polynomial $f_1(x) = 1$ satisfies $\sum_{r=1}^{n}f_1(r) = n$

Mixed exercise 3

Throughout this exercise you may assume the standard results for $\sum_{r=1}^{n}r$, $\sum_{r=1}^{n}r^2$ and $\sum_{r=1}^{n}r^3$.

1 Evaluate:

a $\sum_{r=1}^{10}r$ **b** $\sum_{r=10}^{50}r$ **c** $\sum_{r=1}^{10}r^2$ **d** $\sum_{r=1}^{10}r^3$

e $\sum_{r=26}^{50}r^2$ **f** $\sum_{r=50}^{100}r^3$ **g** $\sum_{r=1}^{60}r + \sum_{r=1}^{60}r^2$

2 Write each of the following as an expression in terms of n.

a $\sum_{r=1}^{n}(3r-5)$ **b** $\sum_{r=1}^{n}(r^2+r)$ **c** $\sum_{r=1}^{n}(3r^2+7r)$

d $\sum_{r=1}^{n}(4r^3+6r^2)$ **e** $\sum_{r=1}^{n}(r^2-2r)$ **f** $\sum_{r=1}^{n}(r^2-3r)$

g $\sum_{r=1}^{n}(r^2-5)$ **h** $\sum_{r=1}^{n}(2r^3+3r^2+r+4)$

E **3** Evaluate $\sum_{r=1}^{30}r(3r-1)$. **(5 marks)**

E **4 a** Show that $\sum_{r=1}^{n}r^2(r-3) = \frac{1}{4}n(n+1)(n^2+an+b)$, where a and b are integers to be found. **(4 marks)**

b Hence evaluate $\sum_{r=1}^{20}r^2(r-3)$. **(2 marks)**

E **5 a** Show that $\sum_{r=1}^{n}(2r-1)^2 = \frac{1}{3}n(an+b)(an-b)$, where a and b are integers to be found. **(5 marks)**

b Hence find $\sum_{r=1}^{2n}(2r-1)^2$. **(2 marks)**

E **6 a** Show that $\sum_{r=1}^{n} r(r+2) = \frac{1}{6}n(n+1)(an+b)$, where a and b are integers to be found. **(4 marks)**

b Hence evaluate $\sum_{r=15}^{30} r(r+2)$. **(3 marks)**

E/P **7 a** Show that $\sum_{r=n+1}^{2n} r^2 = \frac{1}{6}n(2n+1)(an+b)$, where a and b are integers to be found. **(4 marks)**

b Hence evaluate $\sum_{r=16}^{30} r^2$. **(2 marks)**

E/P **8 a** Show that $\sum_{r=1}^{n} (r^2 - r - 1) = \frac{1}{3}n(n^2 - 4)$. **(4 marks)**

b Hence evaluate $\sum_{r=10}^{40} (r^2 - r - 1)$. **(3 marks)**

c Find the value of n such that $\sum_{r=1}^{n} (r^2 - r - 1) = \sum_{r=1}^{2n} r$. **(5 marks)**

E/P **9 a** Show that $\sum_{r=1}^{n} r(2r^2 + 1) = \frac{1}{2}n(n+1)(n^2 + n + 1)$. **(4 marks)**

b Hence show that there are no values of n that satisfy $\sum_{r=1}^{n} r(2r^2 + 1) = \sum_{r=1}^{n} (100r^2 - r)$. **(6 marks)**

E/P **10 a** Show that $\sum_{r=1}^{n} r(r+1)^2 = \frac{1}{12}n(n+1)(n+2)(an+b)$, where a and b are integers to be found. **(5 marks)**

b Hence find the value of n that satisfies $\sum_{r=1}^{n} r(r+1)^2 = \sum_{r=1}^{n} 70r$. **(6 marks)**

E/P **11** Find the value of n that satisfies $\sum_{r=1}^{n} r^2 = \sum_{r=1}^{n+1} (9r + 1)$. **(7 marks)**

Challenge

Show that:

a $\sum_{i=1}^{n}\left(\sum_{r=1}^{i} r^2\right) = \frac{1}{12}n(n+1)^2(n+2)$

b $\sum_{j=1}^{n}\left(\sum_{i=1}^{j}\left(\sum_{r=1}^{i} r\right)\right) = \frac{1}{24}n(n+1)(n+2)(n+3)$

Summary of key points

1 To find the sum of a series of constant terms you can use the formula $\sum_{r=1}^{n} 1 = n$.

2 The formula for the sum of the first n natural numbers is $\sum_{r=1}^{n} r = \frac{1}{2}n(n+1)$.

3 To find the sum of a series that does not start at $r = 1$, use $\sum_{r=k}^{n} f(r) = \sum_{r=1}^{n} f(r) - \sum_{r=1}^{k-1} f(r)$

4 You can rearrange expressions involving sigma notation.

- $\sum_{r=1}^{n} kf(r) = k\sum_{r=1}^{n} f(r)$
- $\sum_{r=1}^{n} (f(r) + g(r)) = \sum_{r=1}^{n} f(r) + \sum_{r=1}^{n} g(r)$

5 The formula for the sum of the squares of the first n natural numbers is

$$\sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1)$$

6 The formula for the sum of the cubes of the first n natural numbers is

$$\sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n+1)^2$$